

How to Read 100 Million Blogs (& Classify Deaths Without Physicians)

Gary King
Harvard University

July 17, 2007

- Daniel Hopkins and Gary King. “Extracting Systematic Social Science Meaning from Text”

- Daniel Hopkins and Gary King. “Extracting Systematic Social Science Meaning from Text”
- Gary King and Ying Lu. “Verbal Autopsy Methods with Multiple Causes of Death,” tentatively to appear, *Statistical Science*

- Daniel Hopkins and Gary King. “Extracting Systematic Social Science Meaning from Text”
- Gary King and Ying Lu. “Verbal Autopsy Methods with Multiple Causes of Death,” tentatively to appear, *Statistical Science*
- Copies at <http://gking.harvard.edu>

Inputs and Target Quantities of Interest

Inputs and Target Quantities of Interest

- Input Data:

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - **individual document classifications** (spam filters)

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - individual document classifications (spam filters)
 - proportion in each category (proportion email which is spam)

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - individual document classifications (spam filters)
 - proportion in each category (proportion email which is spam)
- Estimation

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - **individual document classifications** (spam filters)
 - **proportion in each category** (proportion email which is spam)
- Estimation
 - *Can* get the 2nd by counting the 1st (turns out not to be necessary!)

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - **individual document classifications** (spam filters)
 - **proportion in each category** (proportion email which is spam)
- Estimation
 - *Can* get the 2nd by counting the 1st (turns out not to be necessary!)
 - High classification accuracy $\Rightarrow$ unbiased category proportions

Blogs as a Running Example

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 44–100 million worldwide now.

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 44–100 million worldwide now.
- A democratic technology: 6 million in China and 700,000 in Iran

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 44–100 million worldwide now.
- A democratic technology: 6 million in China and 700,000 in Iran
- “We are living through the largest expansion of expressive capability in the history of the human race”

One specific quantity of interest

One specific quantity of interest

- Affect about President Bush and 2008 candidates

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:
 - Part ordinal, part nominal categorization

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:
 - Part ordinal, part nominal categorization
 - “Sentiment categorization is more difficult than topic classification”

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:
- | <u>Label</u> | <u>Category</u> |
|--------------|----------------------|
| -2 | extremely negative |
| -1 | negative |
| 0 | neutral |
| 1 | positive |
| 2 | extremely positive |
| NA | no opinion expressed |
| NB | not a blog |

- Hard case:

- Part ordinal, part nominal categorization
- “Sentiment categorization is more difficult than topic classification”
- Informal language: “my crunchy gf thinks dubya hid the wmd's, :)!”

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:

- Part ordinal, part nominal categorization
- “Sentiment categorization is more difficult than topic classification”
- Informal language: “my crunchy gf thinks dubya hid the wmd's, :)!”
- Little common internal structure (no inverted pyramid)

The Conversation about John Kerry's Botched Joke

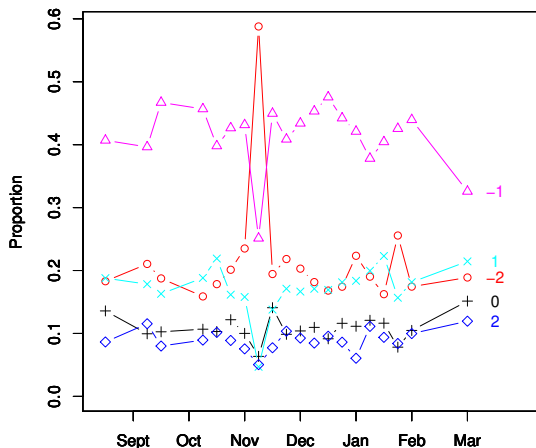
The Conversation about John Kerry's Botched Joke

You know, education — if you make the most of it . . . you can do well. If you don't, you get stuck in Iraq.

The Conversation about John Kerry's Botched Joke

You know, education — if you make the most of it . . . you can do well. If you don't, you get stuck in Iraq.

Affect Towards John Kerry



2006–2007

Representing Text as Numbers

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**
 - Our 10,771 blog posts about Bush and Clinton:
201,676 unigrams, 2,392,027 bigrams, 5,761,979 trigrams.

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**
 - Our 10,771 blog posts about Bush and Clinton:
201,676 unigrams, 2,392,027 bigrams, 5,761,979 trigrams.
 - **keep only** unigrams in $> 1\%$ or $< 99\%$ of documents: 3,672 variables

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**
 - Our 10,771 blog posts about Bush and Clinton: 201,676 unigrams, 2,392,027 bigrams, 5,761,979 trigrams.
 - **keep only** unigrams in $> 1\%$ or $< 99\%$ of documents: 3,672 variables
 - Groups infinite possible posts into “only” $2^{3,672}$ distinct types

Notation

- Document Category

$$D_i = \begin{cases} -2 & \text{extremely negative} \\ -1 & \text{negative} \\ 0 & \text{neutral} \\ 1 & \text{positive} \\ 2 & \text{extremely positive} \\ \text{NA} & \text{no opinion expressed} \\ \text{NB} & \text{not a blog} \end{cases}$$

- Document Category

$$D_i = \begin{cases} -2 & \text{extremely negative} \\ -1 & \text{negative} \\ 0 & \text{neutral} \\ 1 & \text{positive} \\ 2 & \text{extremely positive} \\ \text{NA} & \text{no opinion expressed} \\ \text{NB} & \text{not a blog} \end{cases}$$

- Word Stem Profile:

$$S_i = \begin{cases} S_{i1} = 1 & \text{if "awful" is used, 0 if not} \\ S_{i2} = 1 & \text{if "good" is used, 0 if not} \\ \vdots & \vdots \\ S_{iK} = 1 & \text{if "except" is used, 0 if not} \end{cases}$$

Quantities of Interest

Quantities of Interest

- Computer Science: individual document **classifications**

$$D_1, D_2, \dots, D_L$$

Quantities of Interest

- Computer Science: individual document **classifications**

$$D_1, D_2, \dots, D_L$$

- Social Science: **proportions** in each category

$$P(D) = \begin{pmatrix} P(D = -2) \\ P(D = -1) \\ P(D = 0) \\ P(D = 1) \\ P(D = 2) \\ P(D = \text{NA}) \\ P(D = \text{NB}) \end{pmatrix}$$

Issues with Existing Statistical Approaches

Issues with Existing Statistical Approaches

1 Direct Sampling

Issues with Existing Statistical Approaches

- 1 Direct Sampling
 - Biased without a random sample

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

Issues with Existing Statistical Approaches

- ① Direct Sampling
 - Biased without a random sample
 - nonrandomness common due to population drift, data subdivisions, etc.
 - (Classification of population documents not necessary)
- ② Aggregation of model-based individual classifications

Issues with Existing Statistical Approaches

- ① Direct Sampling
 - Biased without a random sample
 - nonrandomness common due to population drift, data subdivisions, etc.
 - (Classification of population documents not necessary)
- ② Aggregation of model-based individual classifications
 - Biased without a random sample

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless
 - $P(D|\mathbf{S})$ encompasses the “true” model.

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless
 - $P(D|\mathbf{S})$ encompasses the “true” model.
 - $\mathbf{S}$ spans the space of all predictors of D (i.e., all information in the document)

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless
 - $P(D|\mathbf{S})$ encompasses the “true” model.
 - $\mathbf{S}$ spans the space of all predictors of D (i.e., all information in the document)
- Bias even with optimal classification and high % correctly classified

Using Misclassification Rates to Correct Proportions

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**
- **Result:** vastly improved estimates of category proportions

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**
- **Result:** vastly improved estimates of category proportions
- (No new assumptions beyond that of the classifier)

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**
- **Result:** vastly improved estimates of category proportions
- (No new assumptions beyond that of the classifier)
- (still requires random samples, individual classification, etc)

Formalization from Epidemiology

(Levy and Kass, 1970)

Formalization from Epidemiology

(Levy and Kass, 1970)

- Accounting identity for 2 categories:

$$P(\hat{D} = 1) = (\text{sens})P(D = 1) + (1 - \text{spec})P(D = 2)$$

Formalization from Epidemiology

(Levy and Kass, 1970)

- Accounting identity for 2 categories:

$$P(\hat{D} = 1) = (\text{sens})P(D = 1) + (1 - \text{spec})P(D = 2)$$

- Solve:

$$P(D = 1) = \frac{P(\hat{D} = 1) - (1 - \text{spec})}{\text{sens} - (1 - \text{spec})}$$

Formalization from Epidemiology

(Levy and Kass, 1970)

- Accounting identity for 2 categories:

$$P(\hat{D} = 1) = (\text{sens})P(D = 1) + (1 - \text{spec})P(D = 2)$$

- Solve:

$$P(D = 1) = \frac{P(\hat{D} = 1) - (1 - \text{spec})}{\text{sens} - (1 - \text{spec})}$$

- Use this equation to correct $P(\hat{D})$

Generalizations: J Categories, No Individual Classification

(King and Lu, 2007)

Generalizations: J Categories, No Individual Classification

(King and Lu, 2007)

- Accounting identity for J categories

$$P(\hat{D} = j) = \sum_{j'=1}^J P(\hat{D} = j | D = j') P(D = j')$$

Generalizations: J Categories, No Individual Classification

(King and Lu, 2007)

- Accounting identity for J categories

$$P(\hat{D} = j) = \sum_{j'=1}^J P(\hat{D} = j | D = j') P(D = j')$$

- Drop $\hat{D}$ calculation, since $\hat{D} = f(\mathbf{S})$:

$$P(\mathbf{S} = s) = \sum_{j'=1}^J P(\mathbf{S} = s | D = j') P(D = j')$$

Generalizations: J Categories, No Individual Classification

(King and Lu, 2007)

- Accounting identity for J categories

$$P(\hat{D} = j) = \sum_{j'=1}^J P(\hat{D} = j | D = j') P(D = j')$$

- Drop $\hat{D}$ calculation, since $\hat{D} = f(\mathbf{S})$:

$$P(\mathbf{S} = s) = \sum_{j'=1}^J P(\mathbf{S} = s | D = j') P(D = j')$$

- Simplify to an equivalent matrix expression:

$$P(\mathbf{S}) = P(\mathbf{S} | D) P(D)$$

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Document category proportions (quantity of interest)

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Word stem profile proportions (estimate in unlabeled set by tabulation)

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Word stem profiles, by category (estimate in *labeled* set by tabulation)

Estimation

The matrix expression again:

$$\begin{matrix} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{matrix}$$
$$\implies \mathbf{Y} = \mathbf{X}\beta$$

Alternative symbols (to emphasize the linear equation)

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \end{array}$$
$$\implies \mathbf{Y} = \mathbf{X}\beta \implies \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

Solve for quantity of interest (with no error term)

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 & \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 & \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} & & \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies Y = X\beta & \implies & \beta = (X'X)^{-1}X'y \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results
 - Equivalent to kernel density smoothing of sparse categorical data

The matrix expression again:

$$\begin{array}{ccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{array}$$
$$\implies Y = X\beta \quad \implies \quad \beta = (X'X)^{-1}X'y$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results
 - Equivalent to kernel density smoothing of sparse categorical data
 - Use constrained LS to constrain $P(D)$ to simplex

Estimation

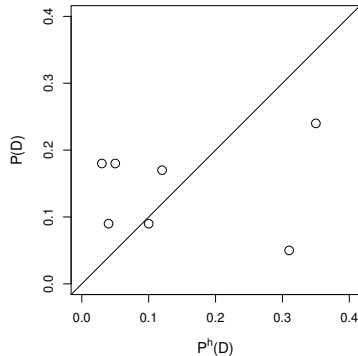
The matrix expression again:

$$\begin{array}{ccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{array}$$
$$\implies Y = X\beta \quad \implies \quad \beta = (X'X)^{-1}X'y$$

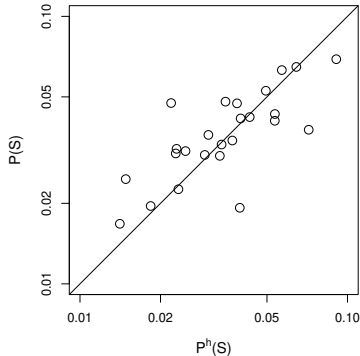
- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results
 - Equivalent to kernel density smoothing of sparse categorical data
 - Use constrained LS to constrain $P(D)$ to simplex
- Uncertainty estimates by bootstrapping

A Nonrandom Hand-coded Sample

**Differences in Document
Category Frequencies**

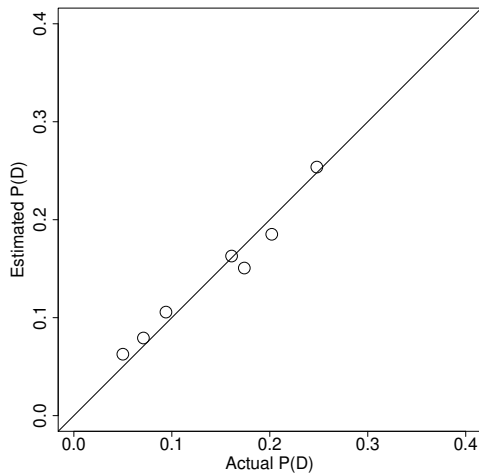


**Differences in Word
Profile Frequencies**

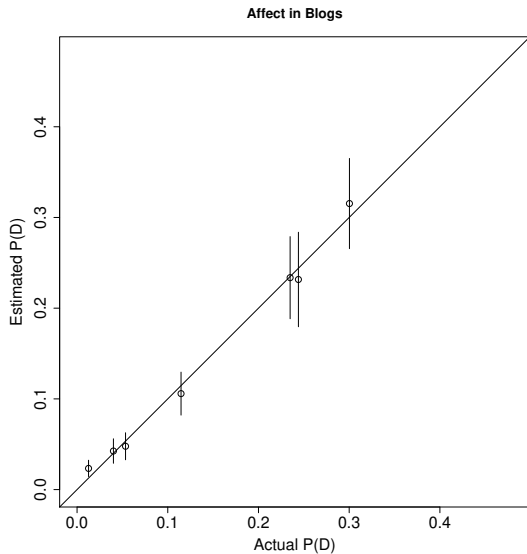


All existing methods would fail with these data.

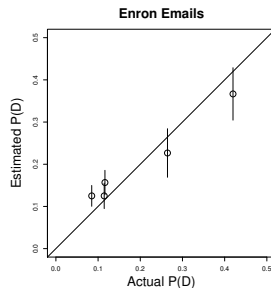
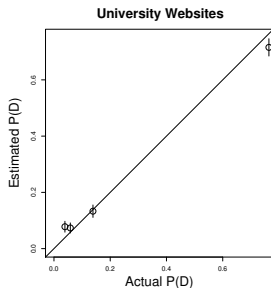
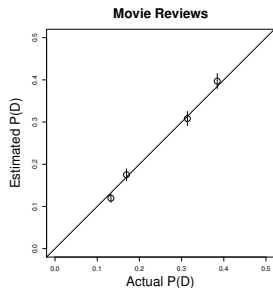
Accurate Estimates



Out of Sample Validation: Blogs



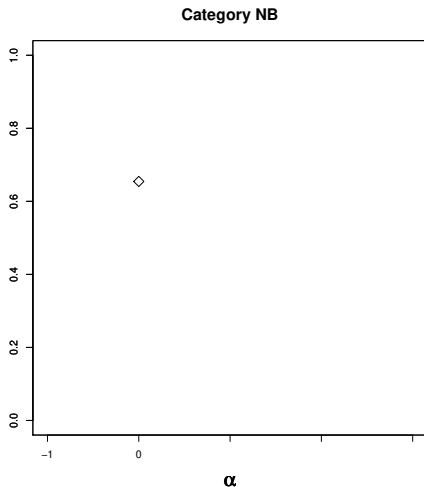
Out of Sample Validation: Other Examples



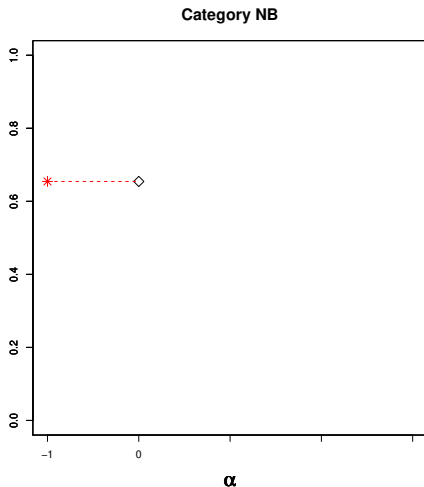
Misclassification Matrix for Blog Posts

	-2	-1	0	1	2	NA	NB	$P(D_1)$
-2	.70	.10	.01	.01	.00	.02	.16	.28
-1	.33	.25	.04	.02	.01	.01	.35	.08
0	.13	.17	.13	.11	.05	.02	.40	.02
1	.07	.06	.08	.20	.25	.01	.34	.03
2	.03	.03	.03	.22	.43	.01	.25	.03
NA	.04	.01	.00	.00	.00	.81	.14	.12
NB	.10	.07	.02	.02	.02	.04	.75	.45

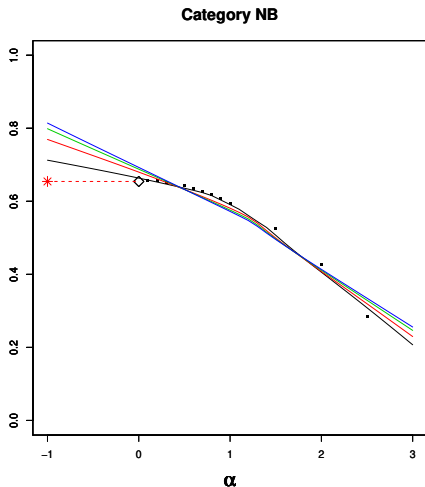
SIMEX Analysis of “Not a Blog” Category



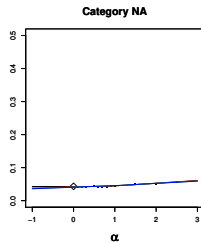
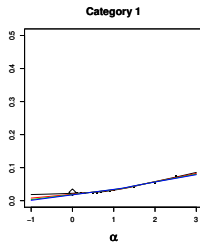
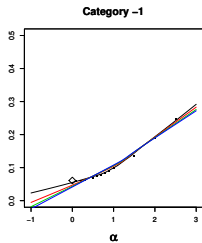
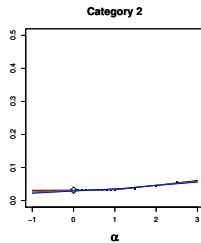
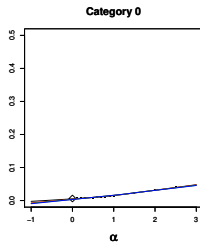
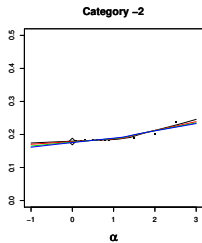
SIMEX Analysis of “Not a Blog” Category



SIMEX Analysis of “Not a Blog” Category



SIMEX Analysis of Other Categories



Verbal Autopsy Methods

Verbal Autopsy Methods

- The Problem

Verbal Autopsy Methods

- The Problem
 - Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

Verbal Autopsy Methods

- The Problem
 - Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
 - High quality death registration: only 23/192 countries
- Existing Approaches

Verbal Autopsy Methods

- The Problem
 - Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
 - High quality death registration: only 23/192 countries
- Existing Approaches
 - Ask relatives or caregivers 50-100 symptom questions

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

- Existing Approaches

- Ask relatives or caregivers 50-100 symptom questions
- Ask physicians to determine cause of death (low intercoder reliability)

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

- Existing Approaches

- Ask relatives or caregivers 50-100 symptom questions
- Ask physicians to determine cause of death (low intercoder reliability)
- Apply expert algorithms (high reliability, low validity)

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

- Existing Approaches

- Ask relatives or caregivers 50-100 symptom questions
- Ask physicians to determine cause of death (low intercoder reliability)
- Apply expert algorithms (high reliability, low validity)
- Find deaths with medically certified causes from a local hospital, trace caregivers to their homes, ask the same symptom questions, and statistically classify deaths in population (model-dependent, low accuracy)

An Alternative Approach

An Alternative Approach

- Document-Category, Cause of Death,

$$D_i = \begin{cases} 1 & \text{if bladder cancer} \\ 2 & \text{if cardiovascular disease} \\ 3 & \text{if transportation accident} \\ \vdots & \vdots \\ J & \text{if infectious respiratory} \end{cases}$$

An Alternative Approach

- Document ~~Category~~, Cause of ~~Death~~,

$$D_i = \begin{cases} 1 & \text{if bladder cancer} \\ 2 & \text{if cardiovascular disease} \\ 3 & \text{if transportation accident} \\ \vdots & \vdots \\ J & \text{if infectious respiratory} \end{cases}$$

- Word ~~Stem~~ Profile, ~~Symptoms~~:

$$S_i = \begin{cases} S_{i1} = 1 & \text{if "breathing difficulties", 0 if not} \\ S_{i2} = 1 & \text{if "stomach ache", 0 if not} \\ \vdots & \vdots \\ S_{iK} = 1 & \text{if "diarrhea", 0 if not} \end{cases}$$

An Alternative Approach

- Document ~~Category~~, Cause of ~~Death~~,

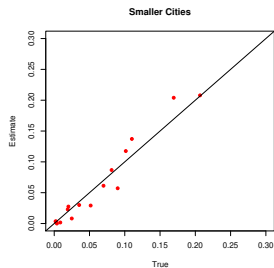
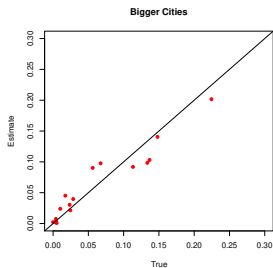
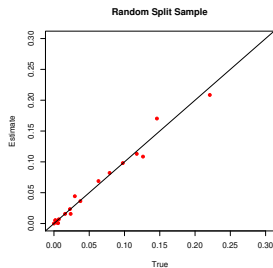
$$D_i = \begin{cases} 1 & \text{if bladder cancer} \\ 2 & \text{if cardiovascular disease} \\ 3 & \text{if transportation accident} \\ \vdots & \vdots \\ J & \text{if infectious respiratory} \end{cases}$$

- Word ~~Stem~~ Profile, ~~Symptoms~~:

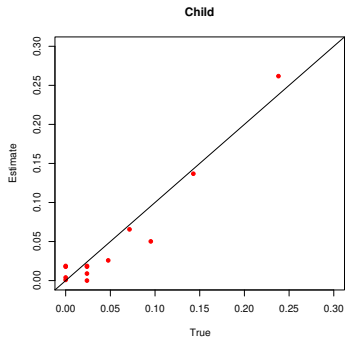
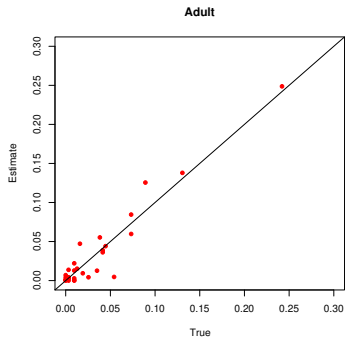
$$S_i = \begin{cases} S_{i1} = 1 & \text{if "breathing difficulties", 0 if not} \\ S_{i2} = 1 & \text{if "stomach ache", 0 if not} \\ \vdots & \vdots \\ S_{iK} = 1 & \text{if "diarrhea", 0 if not} \end{cases}$$

- Apply the **same** methods

Validation in China



Validation in Tanzania



For more information

<http://GKing.Harvard.edu>